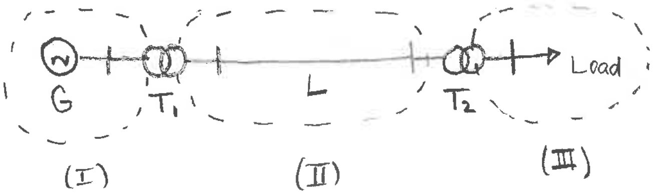
Solution : Per-unit system $S_b = 10\,MVA$



	T_1	T_2
P_{sc} (p.u.) $\rightarrow r_{eq}$	0,01	0,008
P_0 (p.u.) $\rightarrow g_{fe}$	0,003	0,002

Zone I

$\rightarrow V_b = 13{,}2\,kV$

$$Z_b = \frac{(13{,}2k)^2}{10M} = 17{,}424\ \Omega$$

$$I_b = \frac{10M}{\sqrt{3} \times 13{,}2k} = 437{,}387\ A$$

Zone II

$$\rightarrow V_b = 13{,}2k\left(\frac{132}{13{,}2}\right) = 132\,kV$$

$$Z_b = \frac{(132k)^2}{10M} = 1742{,}4\ \Omega$$

$$I_b = \frac{10M}{\sqrt{3} \times 132k} = 43{,}739\ A$$

Zone III

$$\rightarrow V_b = 132k\left(\frac{69}{138}\right) = 66\,kV$$

$$Z_b = \frac{(66k)^2}{10M} = 435{,}6\ \Omega$$

$$I_b = \frac{10M}{\sqrt{3} \times 66k} = 87{,}477\ A$$

Convert values to the new p.u. system

- Generator :

$$X_s = 0.2\left[\frac{(13{,}2k)^2}{5M}\right]\left[\frac{10M}{(13{,}2k)^2}\right] = 0{,}4$$

$$\Rightarrow \boxed{\bar{X}_s = 0{,}4j}$$

- T_1

$$Z_{eq} = \overset{V_{sc}}{(0{,}1)}\left[\frac{(13{,}2k)^2}{5M}\right]\left[\frac{10M}{(13{,}2k)^2}\right] = 0{,}2 \qquad X_{eq} = 0{,}199$$

$$r_{eq} = 0{,}01\left[\text{(same)}\right] = 0{,}02$$

$$Y_T = \overset{I_o}{\cancel{0{,}2}\ 0{,}02}\left[\frac{5M}{(13{,}2k)^2}\right]\left[\frac{(13{,}2k)^2}{10M}\right] = 0{,}01 \qquad b_m = 9{,}887 \times 10^{-3}$$

$$g_{fe} = \cancel{0{,}03}\ 0{,}003\left[\text{same!}\right] = 0{,}0015$$

$$\Rightarrow \boxed{\bar{Z}_{eq} = 0{,}02 + 0{,}199j}$$

$$\Rightarrow \boxed{\bar{Y}_T = 0{,}0015 - 9{,}887 \times 10^{-3} j}$$

- T_2 :

$$Z_{eq} = \overset{V_{sc}}{(0{,}08)}\left[\frac{(138k)^2}{10M}\right]\left[\frac{10M}{(132k)^2}\right] = 0{,}0874$$

$$r_{eq} = 0{,}008\left[\ \text{"}\ \right] = 0{,}0087$$

$\Rightarrow X_{eq} = 0{,}0870$

$$\Rightarrow \boxed{\bar{Z}_{eq} = 0{,}0087 + 0{,}0870j}$$

$$Y_T = \overset{I_o}{(0{,}015)}\left[\frac{10M}{(138k)^2}\right]\left[\frac{(132k)^2}{10M}\right] = 13{,}724 \times 10^{-3}$$

$$g_{fe} = 0{,}002\left[\quad\right] = 1{,}8299 \times 10^{-3}$$

$\Rightarrow b_m = 13{,}601 \times 10^{-3}$

$$\Rightarrow \boxed{Y_T = 1{,}8299 \times 10^{-3} - 13{,}601 \times 10^{-3} j}$$